MIL-STD-1312-23
26 JULY 1985
SUPERSEDING
MIL-STD-1312 (in part)
31 MAY 1967

MILITARY STANDARD

FASTENER TEST METHODS

METHOD 23,

TENSILE STRENGTH

OF PANEL FASTENERS

FSC 53GP

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 23, Tensile Strength of Panel Fasteners

MIL-STD-1312-23

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

This standard sets forth a standard test procedure for determining the tensile strength for all types of quick operating panel fasteners at room temperature.

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1. SCOPE

1.1 Applicability. This method covers a procedure for testing all types of Quick Operating Panel Fasteners in tension at room temperature. This procedure is intended to define the test required to determine the tensile strength of the fastener itself, not the strength of the fastener in any sheet combination.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-45662 Calibration System Requirements

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E4 Load Verification of Testing Machines

ASTM E83 Verification and Classification of Extensometers

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

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4.1.1 Testing machine. The testing machine shall be capable of applying a tensile load at a controllable rate. The calibration system for the machine shall conform to the requirements of MIL-C-45662. Its accuracy shall be verified every 12 months by a method complying with ASTM E4, using a calibration device which shall have been calibrated by the National Bureau of Standards not more than 2 years prior to its use. The yield loads, ultimate loads and structural failure loads of the fasteners tested shall be within the loading range of the testing machine as defined in ASTM E4.

4.1.2 Extension measuring device. The extension measuring device shall be an averaging, differential-transformer extensometer, or equivalent, preferably of the separable type. It shall conform to the requirements of class B-1 in ASTM E83 when used in conjunction with an autographic recorder. The extensometer shall be capable of installation so as to measure either the relative movement between the movable and stationary crossheads of the testing machine or the extension of the fastener only, preferably the latter. Load and extension ranges shall be used which give the initial portion of the load-extension curve a slope between 45 and 60 degrees.

4.1.3 Test fixtures. Typical fixtures are shown in figures 1 and 2. Other types of fixtures may be used provided all of the following specifications are met:

a. Hole size for any fastener shall be the maximum body diameter of the fastener to be tested plus 0.001 inch, with a +0.003 - 0.000 tolerance. Alternately, the hole shall be to the diameter and limits specified by the fastener manufacturer.

b. The hole shall be perpendicular to the fixture surface within $\pm$ 1/2 degree.

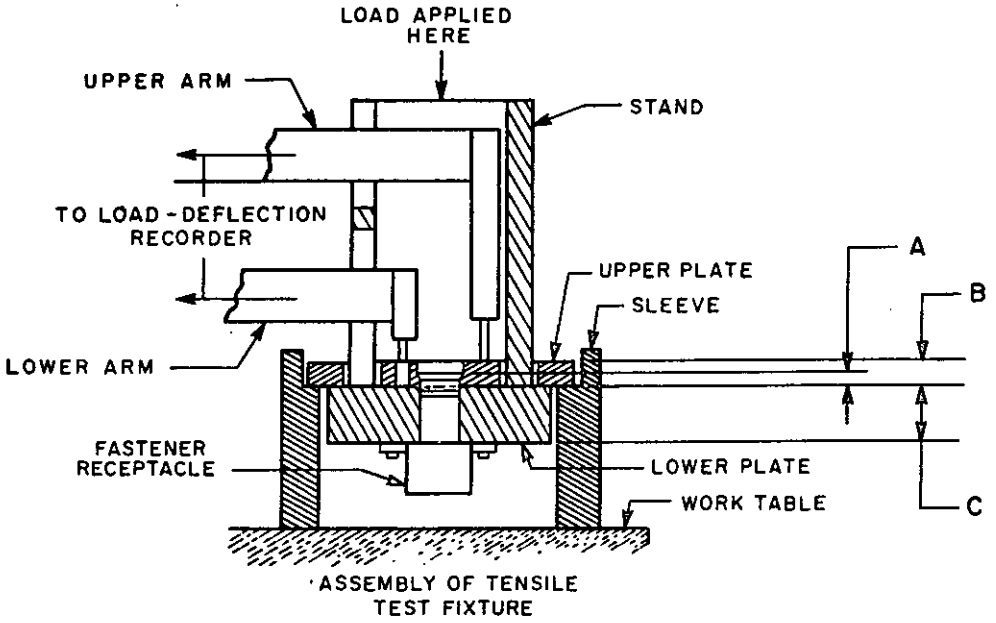
c. Where applicable, the hole shall be chamfered to provide clearance for the head-to-shank fillet of the fastener.

d. The fixture shall be capable of applying an axial tensile load through the centerline of the fastener by means of suitable supporting fixtures.

4.2 Test specimen. The number of specimens to be tested shall be as specified in the procurement document or product specification. The test specimen shall not be altered in any manner other than that described in the product design specification. Fastener grips and plate thicknesses shall be such that the fastener is tested at the maximum grip.

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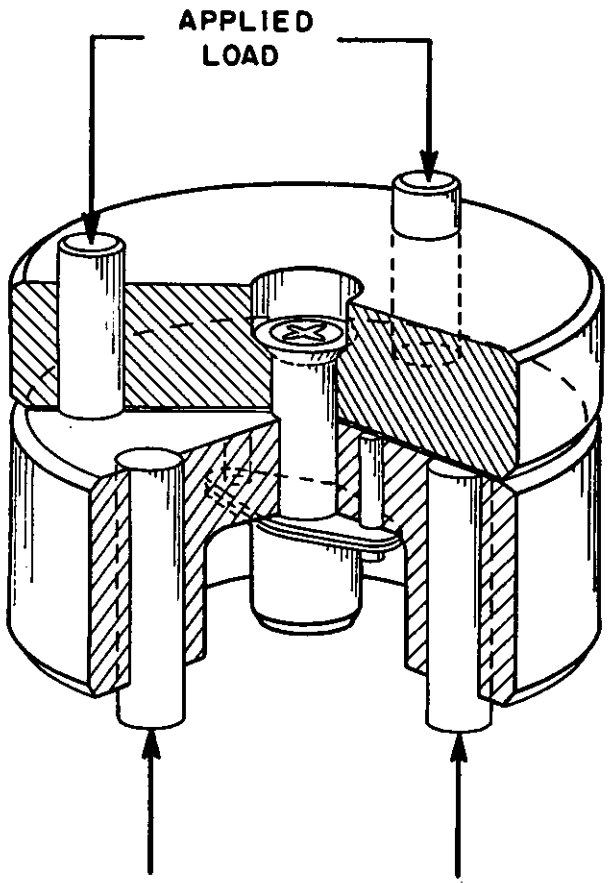
Fastener size dia	Test plate thickness dimension		
	A	B	C
1/4"	0.125	0.250	0.250
5/16"	0.188	0.312	0.375
3/8"	0.250	0.375	0.500



Test plate and fixture material shall be steel plate or bar (MIL-S-18728, MIL-S-5000, or MIL-S-6050) heat treated to 160,000 - 180,000 psi.

FIGURE 1. Typical tension fixture.

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SEE FIGURE 1 FOR
MATERIAL & TEST PLATE
THICKNESS DIMENSIONS

FIGURE 2. Typical tension load.

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5. DETAIL REQUIREMENTS

5.1 Test procedures. The fastener assembly shall be installed in the applicable test fixture and tightened to not more than the maximum locking torque specified for the fasteners being tested. The rated tensile load, as specified in the product design specification, shall be applied thru the fixture in an axial direction tending to pull the fastener apart. This test load shall be applied and released five consecutive times. The fastener shall then be subjected to the locking and unlocking torque tests specified. These torque readings shall be recorded and the fasteners examined for any evidence of failure or permanent deformation.

The fastener shall then be loaded to failure and the ultimate load and description of failure recorded. A load deflection curve shall be constructed.

5.1.1 Load deflection curves. Load deflection curves shall be made by autographic recording. The movement sensing element shall be so installed as to either measure the relative movement between the movable crosshead and the stationary crosshead or to measure the deflection of the fastener only, preferably the latter. The fixture will have the capability of being non-yielding at ultimate load.

5.1.2 Failing load determination. When a structural failure occurs before the ultimate or fracture load, it may be detected on the load-deflection curve as a peak load followed by severe permanent deformation without increase in load or with a decrease in load. A sample load-deflection curve illustrating a "structural failure" is shown on figure 1. The first peak load shall be designated as the "structural failure load" and the highest peak shall be the "ultimate" load.

6. NOTES

6.1 Test reports. The test report shall include the following data:

a. Fastener description.

b. Part number.

c. Lot identification.

d. Manufacturer.

e. Measured body diameters.

f. Grip length.

g. Loading rate or strain rate.

h. Ultimate loads.

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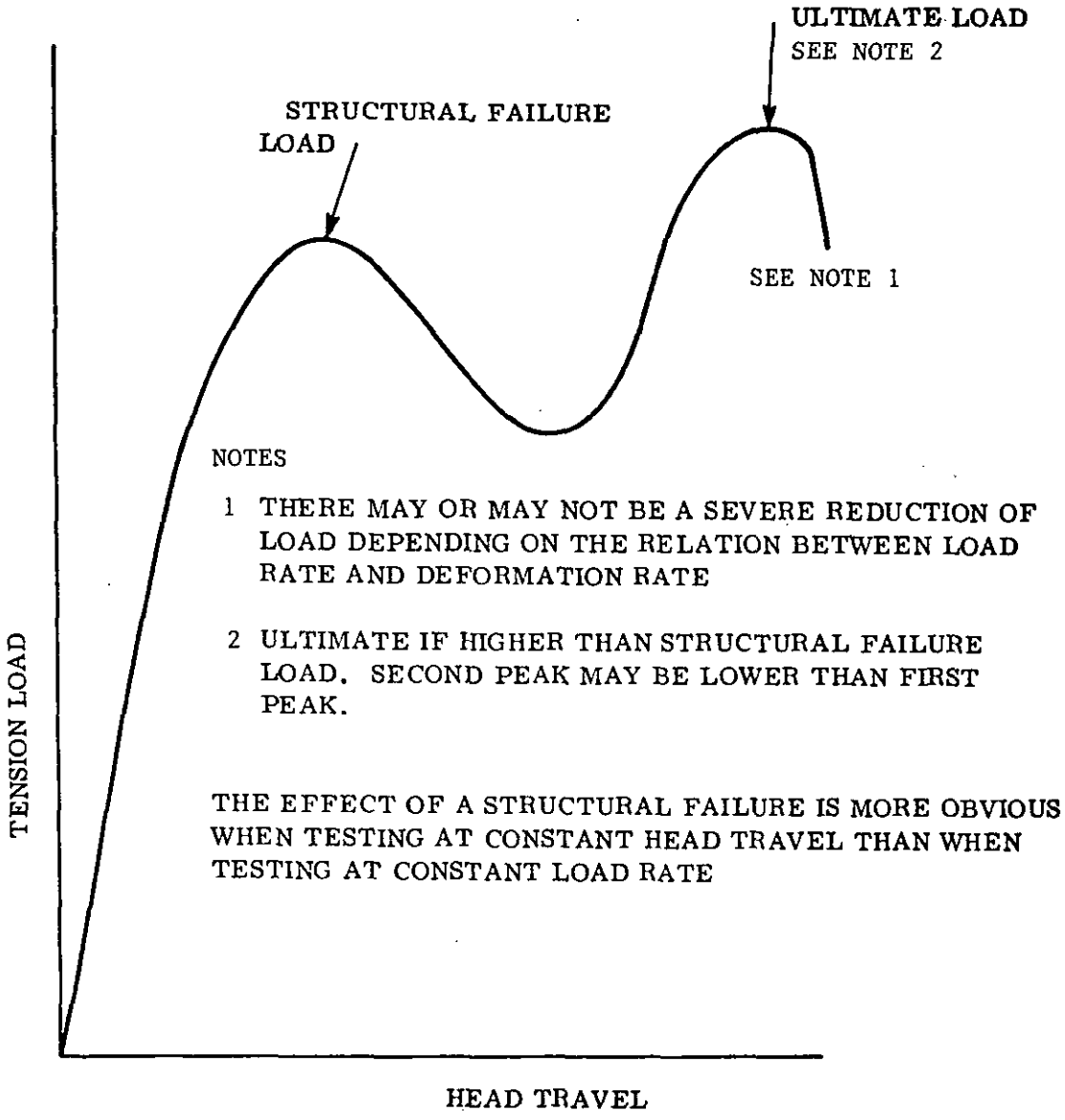


FIGURE 3. Sample load deflection curve.

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i. Structural failure loads.

j. Mode of failure.

k. Fixture used.

l. Load deflection curves.

m. Strain magnification factor.

n. Method of deflection measurement.

Custodians:
Army - AR
Navy - AS
Air Force - 11

Review activities:
Army - AV
Navy - AS, SH
Air Force - 99
DLA - IS

Preparing activity:
Navy - AS
(Project No. 53GP-0143-23)

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MIL-STD-1312-12
26 JULY 1985

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31 MAY 1967

MILITARY STANDARD

FASTENER TEST METHODS

METHOD 12,

THICKNESS OF METALLIC COATINGS

FSC 53GP

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Fastener Test Methods, Method 12, Thickness of Metallic Coating

MIL-STD-1312-12

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FOREWORD

This standard sets forth a standard test procedure for determining the thickness of a metallic coating (plating) on threaded and nonthreaded fasteners.

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1. SCOPE

1.1 Applicability. This test method outlines a standard procedure for determining the thickness of a metallic coating (plating) on threaded and nonthreaded fasteners. This procedure applies to all types of threaded and nonthreaded fasteners and is not limited by configuration or size.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28/6	Screw Thread Standards for Federal Services, Section 6, Gages and Gaging for Unified Screw Threads
FED-STD-151	Metals, Test Methods

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Drop test method. The apparatus shown in Figure of Method of FED-STD-151 illustrates one method of applying the corroding medium. Other types of apparatus may be used if the drops are applied in conformance with Method of FED-STD-151.

4.1.2 Magnetic test method (Magne-Gage or equivalent). Four different magnets and seven different calibration curves are used for various materials and thickness ranges. Table I of Method 522.1 of FED-STD-151, titled "Magnetic and calibration curves for use with the standard Magne-Gage," specifies the proper combination of magnets and calibration curve to use for different coatings and thickness ranges.

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4.1.3 Eddy-current method. Coating thickness shall be determined with an electromagnetic instrument that measures the changes in apparent impedance of the coil inducing the eddy currents into the basis metal. The design of the instrument shall be such that variations in apparent impedance, produced by variations in coil to base metal spacing, can be calibrated to indicate the thickness of the coatings (see 5.1.3).

4.1.4 Microscopic method. Coating thickness shall be determined at a magnification of 400X or more, using a suitable metallographic microscope and a filar micrometer ocular. The ocular shall be calibrated to determine the value of the unit scale division against a scale so graduated that the error will not exceed 2.0 percent. An additional coating shall be applied to the specimen prior to mounting to support the metal coating and assure accurate readings of metal coating thickness.

4.1.5 Dimensional change method. Functional diameter-indicating type thread gages with gaging elements which in length approximate the width of the applicable "Go" thread ring gage, and which have been checked or set with master thread-setting plug gages conforming to FED-STD-H28/6 shall be used.

4.1.6 Anodic dissolution method. An electronic thickness tester (e.g., Kokour) and accessory equipment shall be used.

4.1.7 Strip and weigh method. Coating thickness shall be determined, using a laboratory balance capable of reading to an accuracy of ±0.0001 grams, by the differential of the weights plated and unplated.

4.2 Test specimens.

4.2.1 Location of plate thickness measurement (externally threaded fasteners). The measurement of the thickness of coating shall be made at the locations shown on figure 1. For fasteners having a body, the location of measurement shall be at a point adjacent to the thread runout area. For fasteners threaded to the head, the measurement shall be made on the head peripheral surface when the head configuration permits an accurate measurement determination, or on the thread pitch diameter at an approximate distance of 2D as measured from the point end, or a distance of one-half the length of the fastener when the length is less than 4D.

4.2.2 Nuts. The plating thickness of nuts shall be determined on the bearing surface of the nut at the mid-radius as shown on figure 2.

4.2.3 Fasteners and fastener components of the hollow rivet type. The plating thickness on parts with a length to outside diameter ratio of less than 4 shall be measured on the exterior surface at a point midway between the ends of the part. For parts having a length to

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outside diameter ratio of 4 or more, plating thickness shall be determined on the exterior surface at a location of one D from either end. Plating thickness locations for these parts are shown on figure 2.

5. DETAIL REQUIREMENTS

5.1 Test procedures. When specified, tests for determining thickness of metallic coatings shall be performed in accordance with one of the following methods. The selection of method shall be at the option of the producer. The microscopic method (see 5.1.4) shall be used as the referee method.

5.1.1 Drop test method. The procedure specified in Method of FED-STD-151, shall be followed, with the exception that the test solutions shall be in accordance with the applicable plating processing specification. The plating thickness shall be determined from a time-temperature plating thickness table shown in the applicable plating specification.

5.1.2 Magnetic test method. The procedure shall be performed in accordance with Method 522.1 of FED-STD-151.

5.1.3 Eddy-current method. The procedure shall be performed in accordance with Method 520.1 of FED-STD-151.

5.1.4 Microscopic method. The procedure for the preparation of specimens and measurement of plate thickness shall be as outlined in Method of FED-STD-151 Microscopial Test. The magnification shall be as specified in 4.1.4.

5.1.5 Dimensional change method.

5.1.5.1 The procedure for determining the coating thickness on the external thread of a 60-degree triangular thread form shall be accomplished in the following successive manner:

a. The functional diameters of the coated specimens shall be determined by measuring at three approximate equidistant points around the O.D. of the location areas shown on figure 1. The functional diameter numerical values shall be recorded and averaged for each specimen.

b. The coating shall be removed by chemical solution from the threads of the specimen.

c. The specimens shall be rinsed and dried.

d. Measure and record, as described in a above, the functional thread diameters of the stripped specimens.

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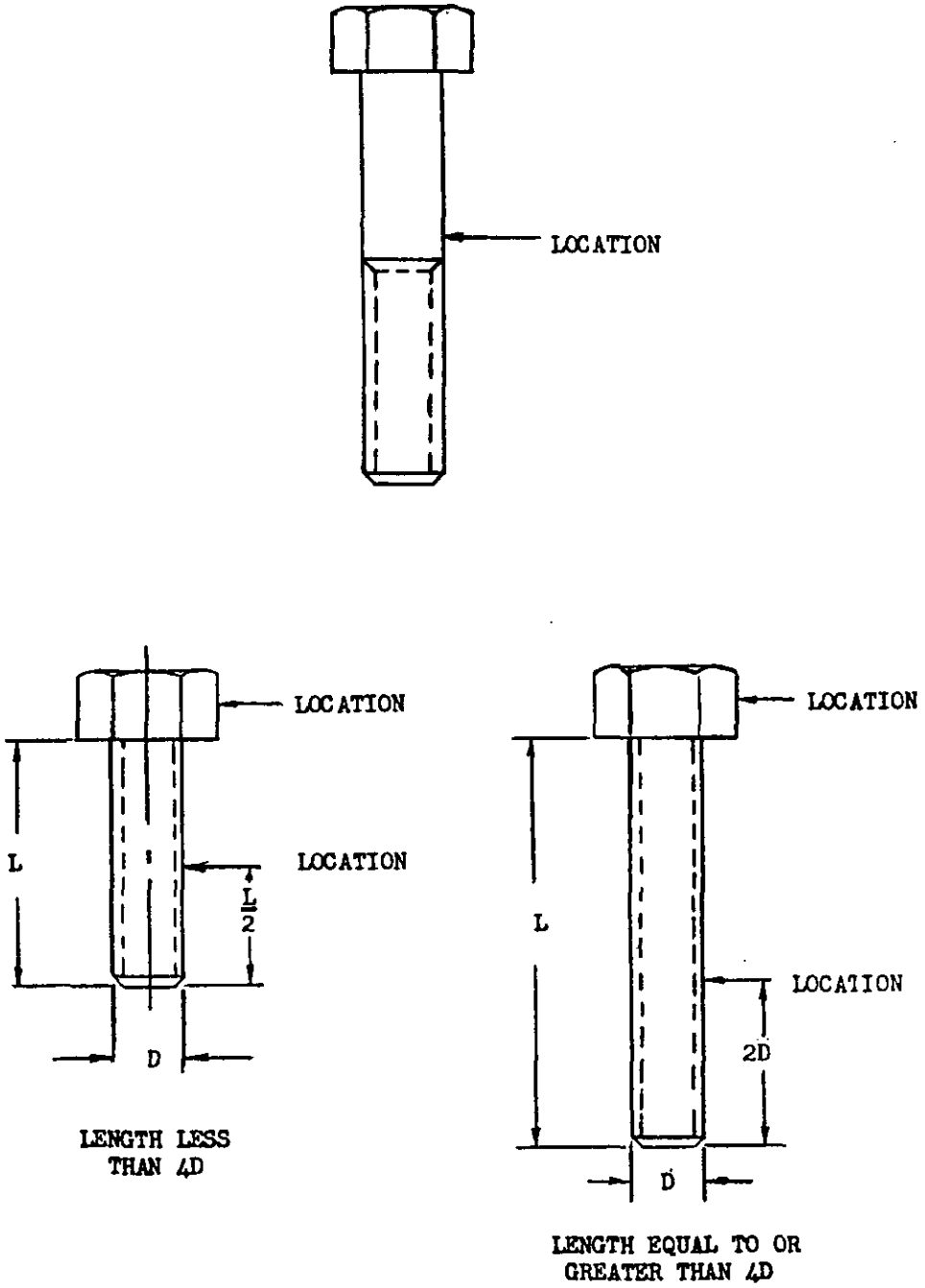


FIGURE 1. Externally threaded fasteners.

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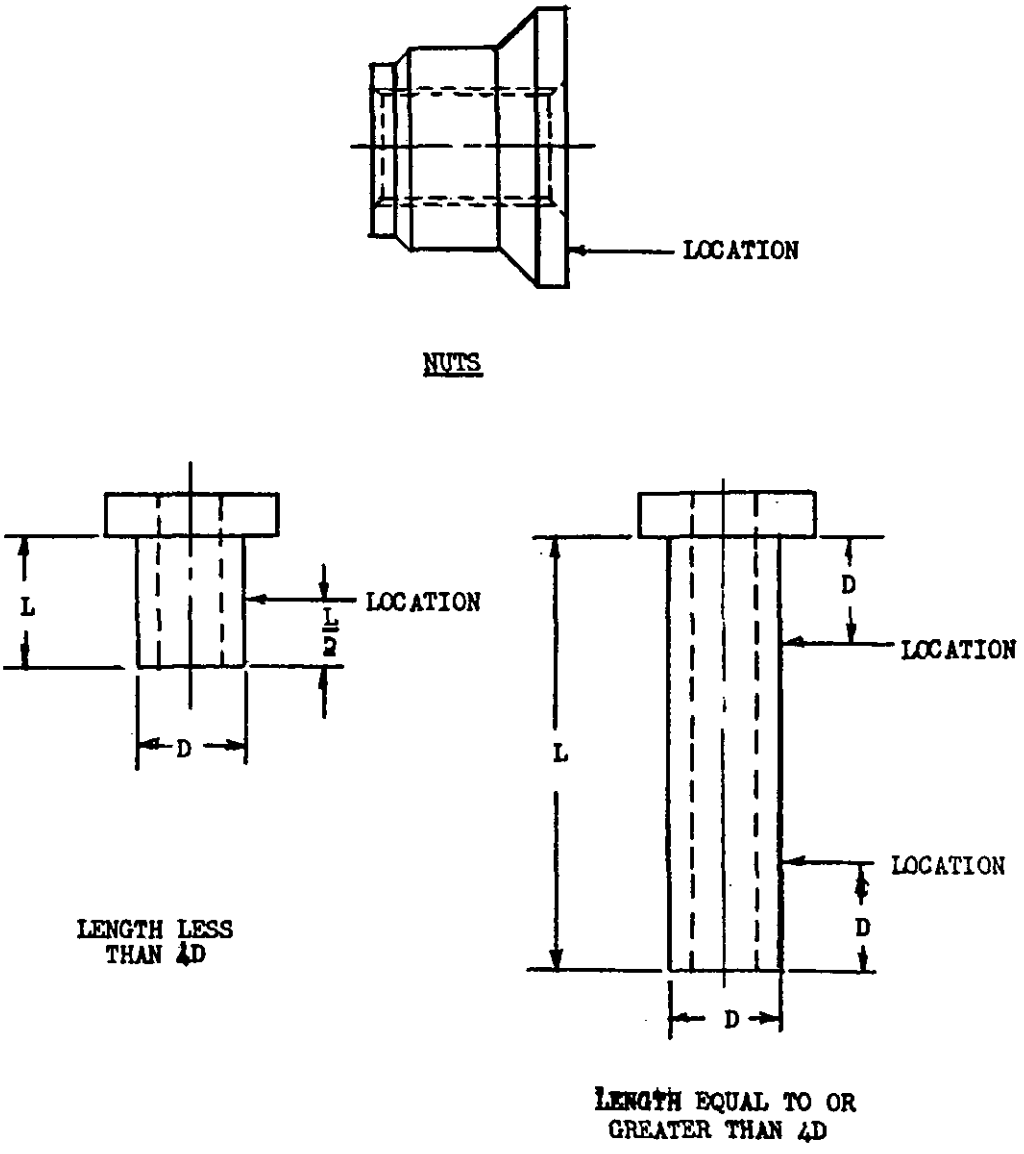


FIGURE 2. Hollow rivets and fastener components.

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e. The approximate coating thickness equals one-quarter of the difference between the average coated and uncoated functional diameter measurements.

5.1.5.2 The procedure for determining the coating thickness on cylindrical or parallel sections shall be accomplished in the following successive manner:

a. Measure and record the coated dimensions.

b. Strip coating.

c. Rinse and dry the parts.

d. Measure and record the uncoated dimensions.

e. Thickness equals one-half the difference between the coated and uncoated measurements.

5.1.6 Strip and weigh method.

5.1.6.1 The procedure for determining the thickness of coating shall be as follows:

a. Weigh the coated fastener.

b. Strip coating from fastener.

c. Rinse and dry parts.

d. Weigh the uncoated fastener.

e. Calculate the approximate thickness of the coating as specified in 5.1.6.2.

NOTE: When multiple platings (such as silver and nickel) are applied, then steps a through d should be repeated for each plating.

5.1.6.2 The procedure for converting weight of plating to thickness shall be as follows:

a. Accurately calculate the surface area of the fastener, treating external and internal areas separately. Internal areas are those that cannot contact each other on two adjacent parts, such as holes, recesses and internal threads.

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b. Calculate thickness from:

$$T = W/K_E A_E + K_I A_I$$

where:

W = weight of plating in grams.
T = thickness of deposit in inches.
A_E = external area in square inches.
A_I = internal area in square inches.
K_E & K_I = constants for external and internal areas for each metallic plating.

Values of K_E and K_I for some commonly deposited metals are as follows:

Metal	K_E	K_I
Zinc	117.0	87.8
Cadmium	141.7	106.3
Copper	147.5	110.6
Nickel	146.0	48.2
Silver	172.5	129.4

6. NOTES

6.1 Test report. The test report shall contain the following data:

a. Fastener description.

b. Lot identification.

c. Material.

d. Test method used for determining plate thickness.

e. Part number.

f. Manufacturer.

g. Plating specification and type.

h. Test results.

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Army - AR
Navy - AS
Air Force - 11

Review activities:
Army - AV
Navy - AS, SH
Air Force - 99
DLA - IS

Preparing activity:
Navy - AS
(Project No. 53GP-0143-12)

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